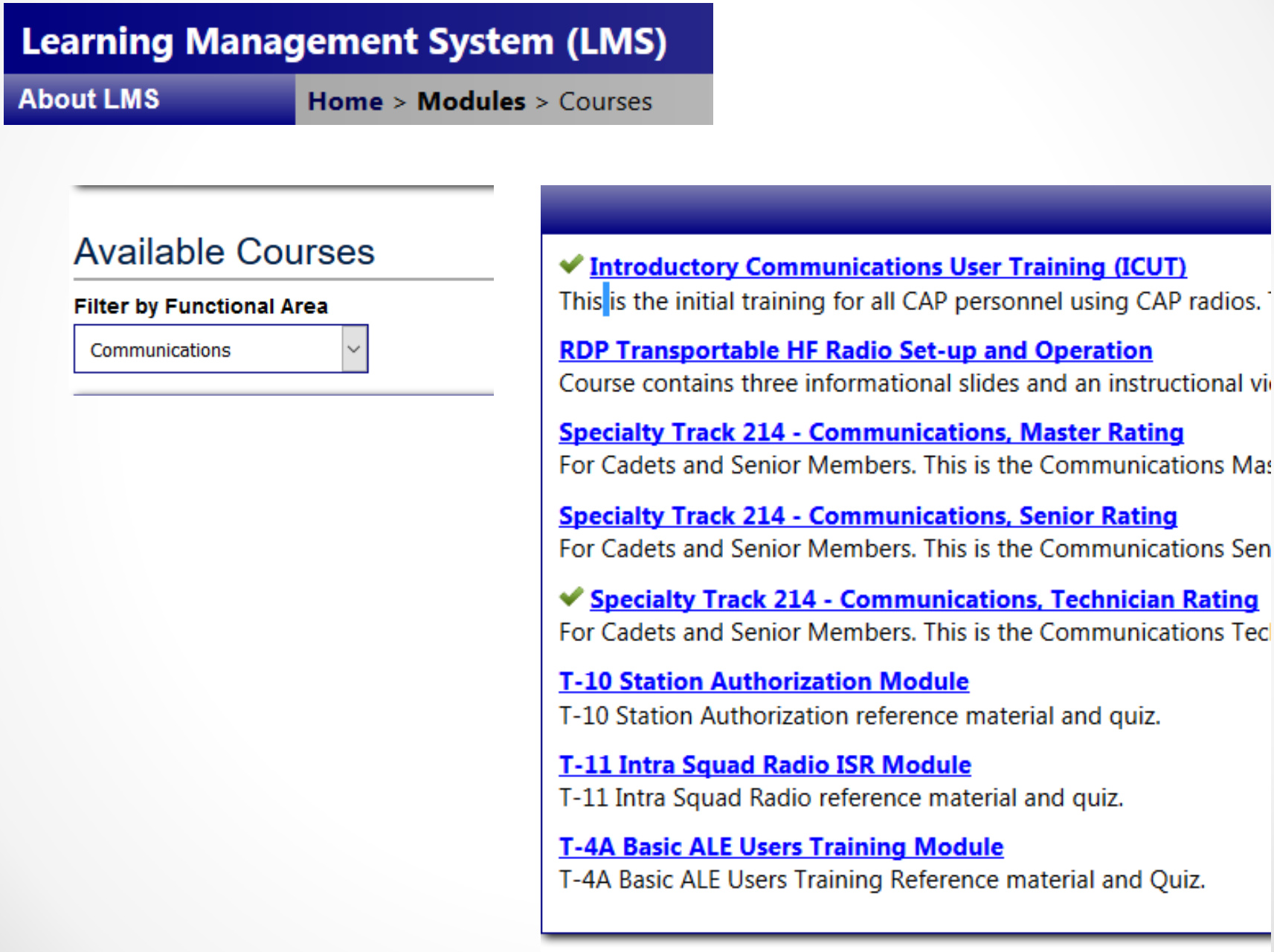
Communications

TRAINING



Introductory User Communications Training – Review (ICUT)

Available Training Resources



Overview

This is a review / overview of the ICUT radio operators requirement and practical evaluation

- Rules and Regulations
- CAP Communications Systems
- Radio Operations
- Radio Procedures
- Message Protocols
- Do's and Don'ts

Rules and Regulations

All CAP Radio Operators shall be familiar with and comply with all applicable CAP Rules and Regulations. The primary CAP Regulations are:

- ✓ CAPR 100-1, “CAP Communications”
- ✓ CAPR 100-3, “Radiotelephone Operations”

For Mission Radio Operators, also refer to the appropriate Specialty Qualification Training Records (SQTR).

CAP Radio Operator Authorization

- Communications Orientation Class Requirements
  - Any member can obtain a authorization to use the CAP radio system by completing the

    Introductory Communications User Training – ICUT

    Found At eServices – Learning Mgmt System – ICUT

    Consists of 11 on-line modules plus a required Skills Evaluation by a SET evaluator

- Two Mission Qualifications are available for missions
  - MRO – Mission Radio Operator
  - CUL – Communications Unit Leader

What's The Role?

Get the message through

CAP radio systems are to operate in times when all other methods of communication fail

Dispite --

Disasters, power outages, cell system failures, internet outage, land line phone systems

Civil Air Patrol Communication Systems

Short Range Communications
- ❖Ultra-High Frequency (UHF)
  - ➢Inter-Squad Radios (ISR)

Medium Range Communications
- ❖Very-High Frequency (VHF)
  - ➢Hand-Held VHF-FM Radio
  - ➢Hand-Held VHF-AM Radio
  - ➢Mobile VHF-FM Radio
  - ➢Mobile VHF-AM Radio
  - ➢Airborne VHF-FM radio
  - ➢Airborne VHF-AM Radio

Long Range Communications
- ❖Very-High Frequency (VHF)
- ❖High Frequency (HF)
  - ➢VHF-FM Airmobile
  - ➢VHF-FM Repeaters
  - ➢HF-SSB (Single Side Band)

4 Systems Utilized today

Plus 1 more

CAP Communications also includes:

- Telephones - Landline and cellular telephones can be used in addition to radio communications.

- INTERNET - E-mail communications, information web pages, internet phone and other methods of communication over the internet.

- GOAL - To have a readily available and comprehensive communications network using a variety of assets.

Radio Operation

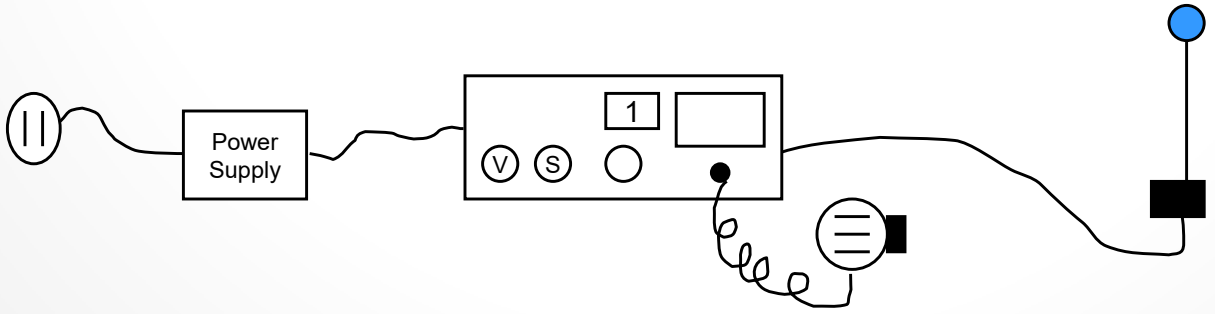
Specific Radio Information will come later.

Common Controls:

- **Power / Volume**
- **Squelch (May or may not be adjustable)**
- **Channel Selector**
- **Mike with Push to Talk Switch (Release to Listen)**

Radio Setup

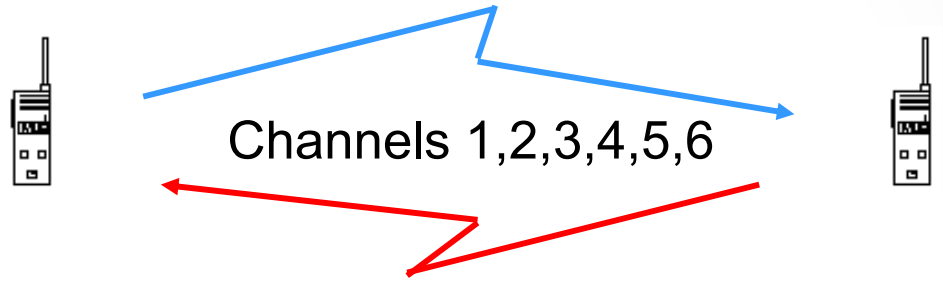
- **Radio Transceiver (VHF-FM, HF-SSB, SAR)**
- **Power Supply (110 VAC or 12 Volt DC, 24 volt in aircraft)**
- **Antenna (Vertical, Magnetic Mount, Dipole)**



Voice Operating Modes

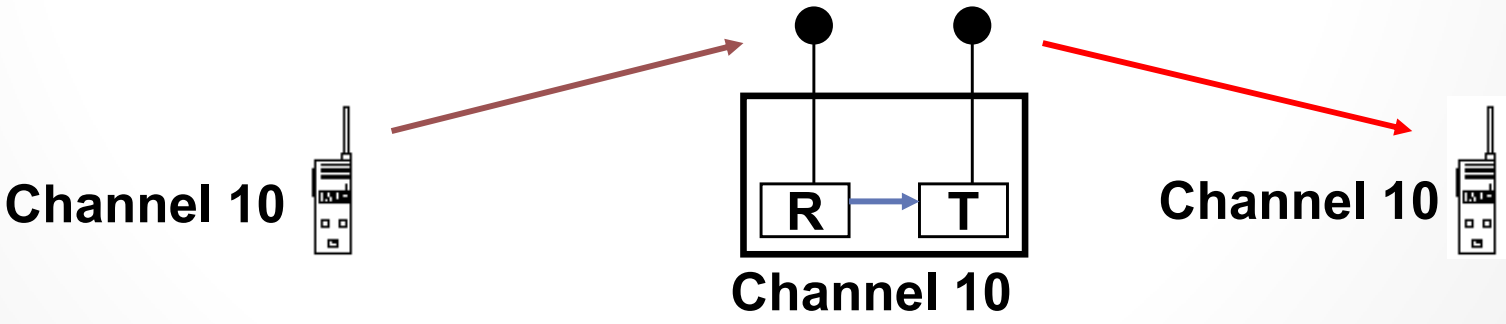
SIMPLEX

Single Frequency - One Station at a Time



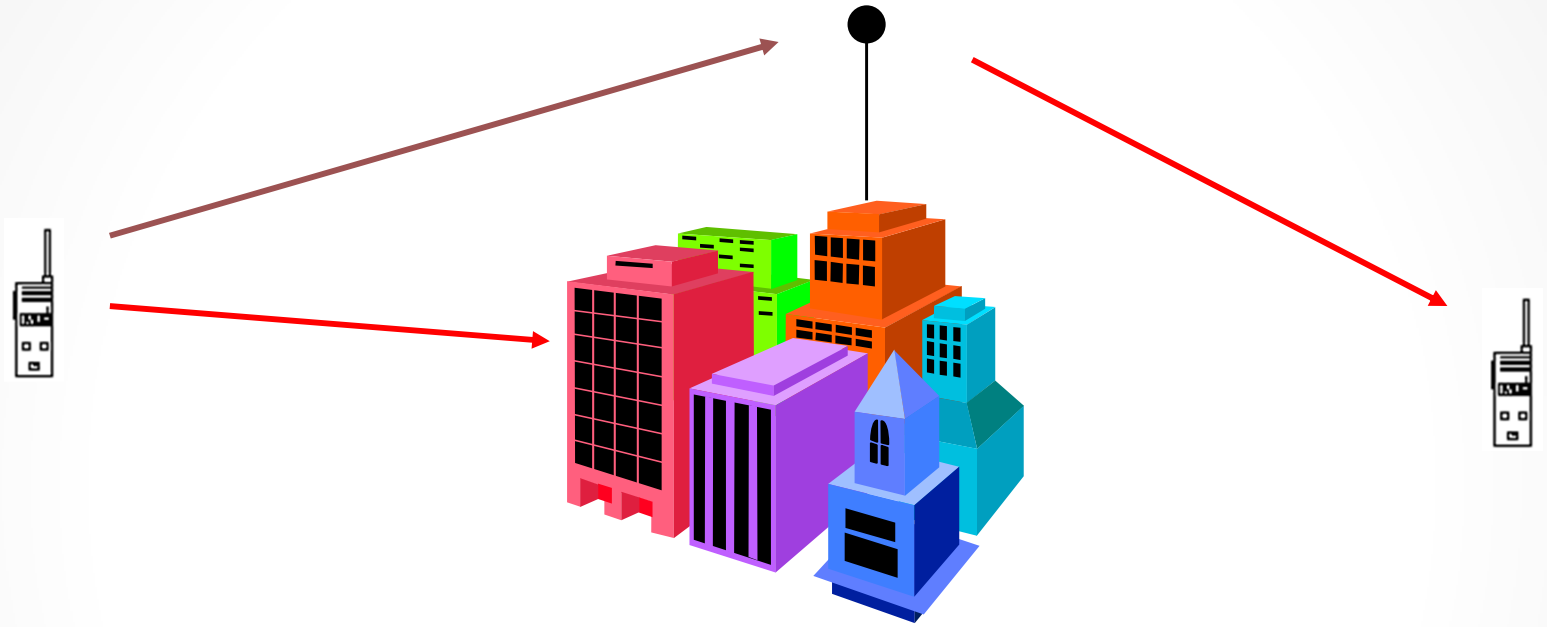
REPEATER

Two Frequencies - One Station at a Time



A "Channel" consist of a transmit frequency, a receive frequency, and sub-audible tones that are not normally heard.

Repeater Operation



A repeater increases the range of land stations due to its high profile location.

Airborne repeaters function the same way.

Radio Operation

CAP has 6 simplex VHF-FM channels:

- **CC-1 (Command Channel)**
- **CC-2**
- **Air 1**
- **Air 2**
- **Guard1 or “Guard”**
- **TAC 1 or Tactical**

Plus the repeater frequencies.

Types of Functional Call Signs

- **MISSION BASE.**
  West Houston Mission Base

- **Ground Team 1, UDF Team 1, etc.**

- **Flight Line**

- **IC, Safety, Operations, Com, Medical, etc**

- **HIGHBIRD**

Standard call signs
CAP0137 (KIWS van)

Radio Operator Responsibilities

For both Send & Receives

- **GET THE MESSAGE THROUGH**

6 Habits of a Good Radio Operator

- Listen
- Speak clearly
- Annunciate your words.
- Speak slowly
- Remain calm no matter what happens - Never Panic
- THINK - “Use Your Head”

Message Protocol

All messages have 4 parts

- Who I Am Calling
- Who I Am
- The Message
- Termination

Message Protocol

Initial Contact Example

- Who I Am Calling
  - KIWS Mission base
- Who I Am
  - This is CAP0137
- The Message
- Termination
  - Over

Example with Information

- Who I Am Calling
  - KIWS Mission base
- Who I Am
  - This is CAP0137
- The Message
  - I have located the target, coordinates to follow
- Termination
  - Over

Transmission Protocol

Typical Message Transmission

- **Listen for "clear" frequency**
- **Establish Contact**
- **Wait For Acknowledgement**
- **Send The Message**
- **Confirm Message Was Received**
- **Release the frequency**

Transmission Protocol

Typical Message Transmission

- **Listen for "clear" frequency**
- **Establish Contact**
  - **KIWS Mission base this is Ground Team 1 "over"**
- **Wait For Acknowledgement**
  - **GT1 this is Mission Base "over"**
- **Send The Message**
  - **Mission base this is GT1, we have located the target "over"**
- **Confirm Message Was Received**
  - **GT1 this is mission base, understood you have located the target "over"**
- **Release the frequency**
  - **GT1 "out"**

You are not done until the message receipt has been confirmed

Reference: 100-3

PROWORDS

Prowords are a special set of words used for clarity and brevity in communications.

Some of the most commonly used prowords are:

• THIS IS	Preface to your call sign
• ROGER	Last transmission received OK
• OVER	I'm done, go ahead
• OUT	I'm done, bye
• WAIT	I will be back in a few seconds
• SAY AGAIN	Say that again
• CORRECTION	Oops! I really meant to say
• WILCO	ROGER and I will comply
• AFFIRMATIVE	Yes

Distress and Emergency Signals

MAYDAY	Distress	**Supercedes all Priority or Routine Traffic**
PAN	Urgency	
SECURITE	Safety	

Repeat each word 3 times:
"PAN PAN PAN this is …"

Phonetic Alphabet

A	Alpha	N	November
B	Bravo	O	Oscar
C	Charlie	P	Papa
D	Delta	Q	Quebec
E	Echo	R	Romeo
F	Foxtrot	S	Sierra
G	Golf	T	Tango
H	Hotel	U	Uniform
I	India	V	Victor
J	Juliet	W	Whiskey
K	Kilo	X	X-Ray
L	Lima	Y	Yankee
M	Mike	Z	Zulu

Numbers

Pronunciation

1WUN

2TOO

3TREE

4FO-WER

5FIFE

6SIX

7SEVEN

8ATE

9NINER

0ZERO

When writing the numbers, do not write them down the way they are pronounced.

For example, do not write "one" as "wun" or "five" as "fife." Write them as "one" and "five."

I SPELL / FIGURES / INITIALS

- Use “I SPELL” for pronounceable words
  --- PIZZA “I SPELL PIZZA PAPA INDIA ZULU ZULU ALPHA PIZZA”

- Use “FIGURE(S)” AND “INITIAL(S)” for non-words
  --- N516F “INITIAL NOVEMBER FIGURES FIVE ONE SIX INTIAL FOXTROT”

CORRECTION

- **Use proword “CORRECTION” to correct a mistake**

Example:

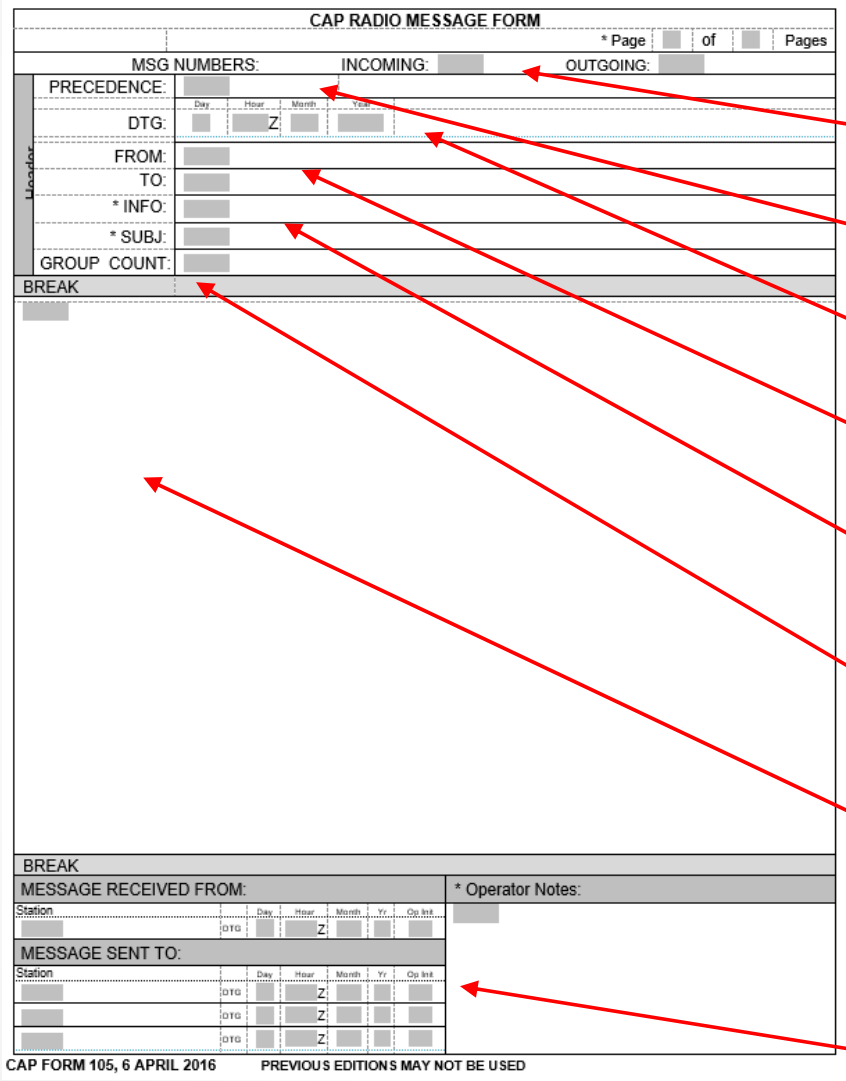
“… Turn right at next corner … CORRECTION Turn left at next corner…”

SENDING NUMBERS

- Prowords “FIGURES”, “DECIMAL”, “TIME”, “INITIALS”

Digit-by-Digit Not “Seven Fifty”	**750**	**“FIGURES SEVEN FIVE ZERO”**
Niner Not Nine	**849**	**“FIGURES EIGHT FOUR NINER”**
Decimal Point	**14.5**	**“FIGURES ONE FOUR DECIMAL FIVE”**
Z Time	**1635Z**	**“TIME ONE SIX THREE FIVE ZULU”**
Initial And Figures	**E21**	**“INITIAL ECHO FIGURES TWO ONE”**
One Figure and Initial	**3-A**	**“FIGURE THREE DASH INITIAL ALPHA”**

CapF105 – Formal Msg Form


CAP RADIO MESSAGE FORM

* Page of Pages

MSG NUMBERS: INCOMING: OUTGOING:

Header

PRECEDENCE:

DTG: Day Hour Z Month Year

FROM:

TO:

* INFO:

* SUBJ:

GROUP COUNT:

BREAK

BREAK

MESSAGE RECEIVED FROM:

Station DTG Day Hour Z Month Yr Op Init

MESSAGE SENT TO:

Station DTG Day Hour Z Month Yr Op Init

* Operator Notes:

CAP FORM 105, 6 APRIL 2016 PREVIOUS EDITIONS MAY NOT BE USED

Key Parts

Msg #

Precedence

DateTimeGroup

To/From

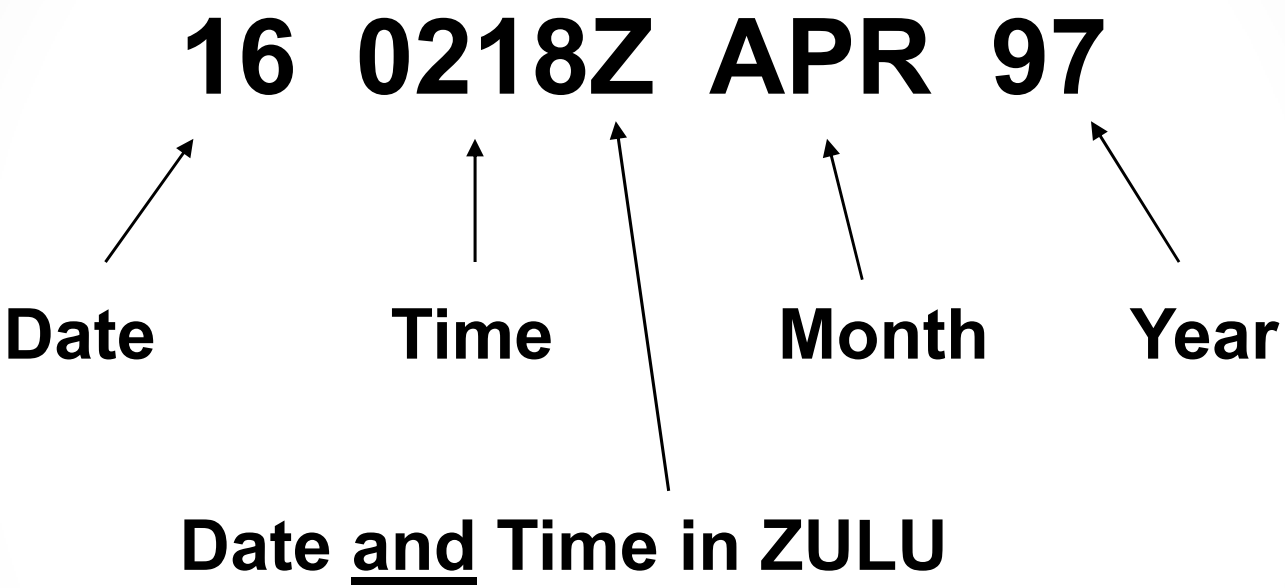
Info/Subject

Group Count

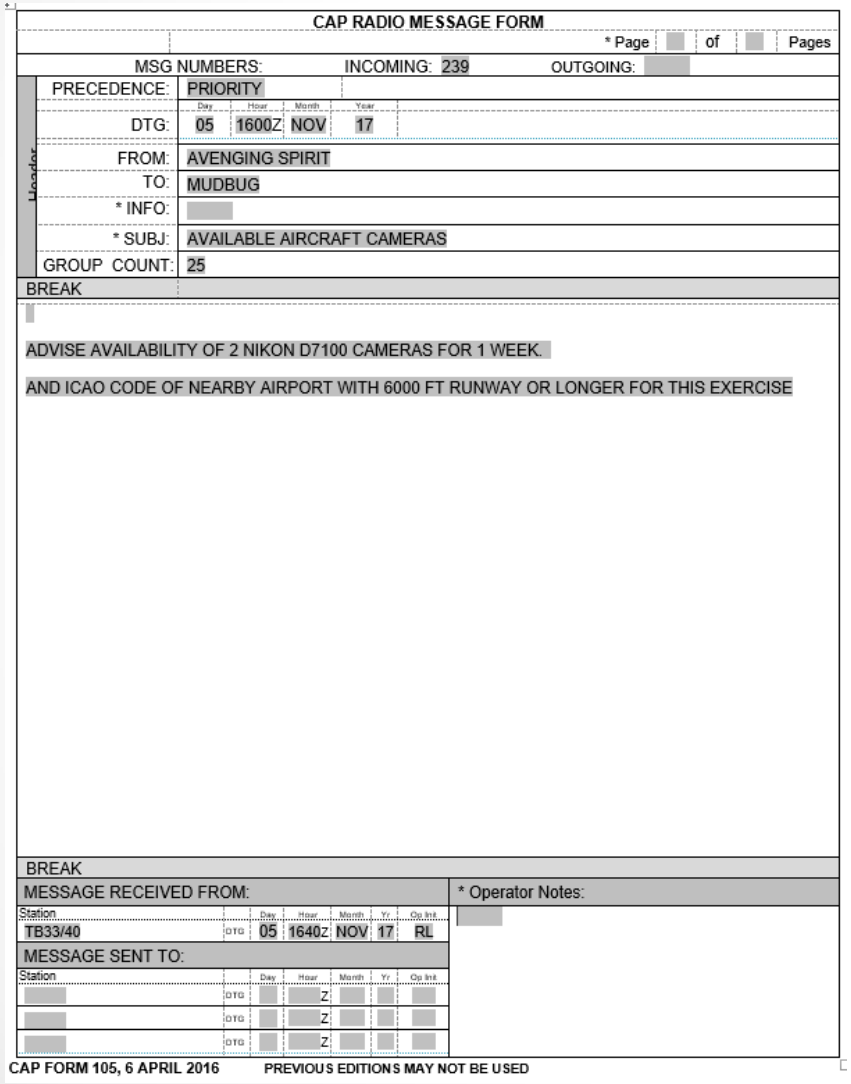
Msg

Rcvd From / Sent To documentation box

Date-Time Group



CapF105 – Formal Msg Form



CAP RADIO MESSAGE FORM

* Page of Pages

MSG NUMBERS: INCOMING: 239 OUTGOING:

Header

PRECEDENCE:	PRIORITY
DTG:	Day 05 Hour 1600Z Month NOV Year 17
FROM:	AVENGING SPIRIT
TO:	MUDBUG
* INFO:	
* SUBJ:	AVAILABLE AIRCRAFT CAMERAS
GROUP COUNT:	25

BREAK

ADVISE AVAILABILITY OF 2 NIKON D7100 CAMERAS FOR 1 WEEK.

AND ICAO CODE OF NEARBY AIRPORT WITH 6000 FT RUNWAY OR LONGER FOR THIS EXERCISE

BREAK

MESSAGE RECEIVED FROM:

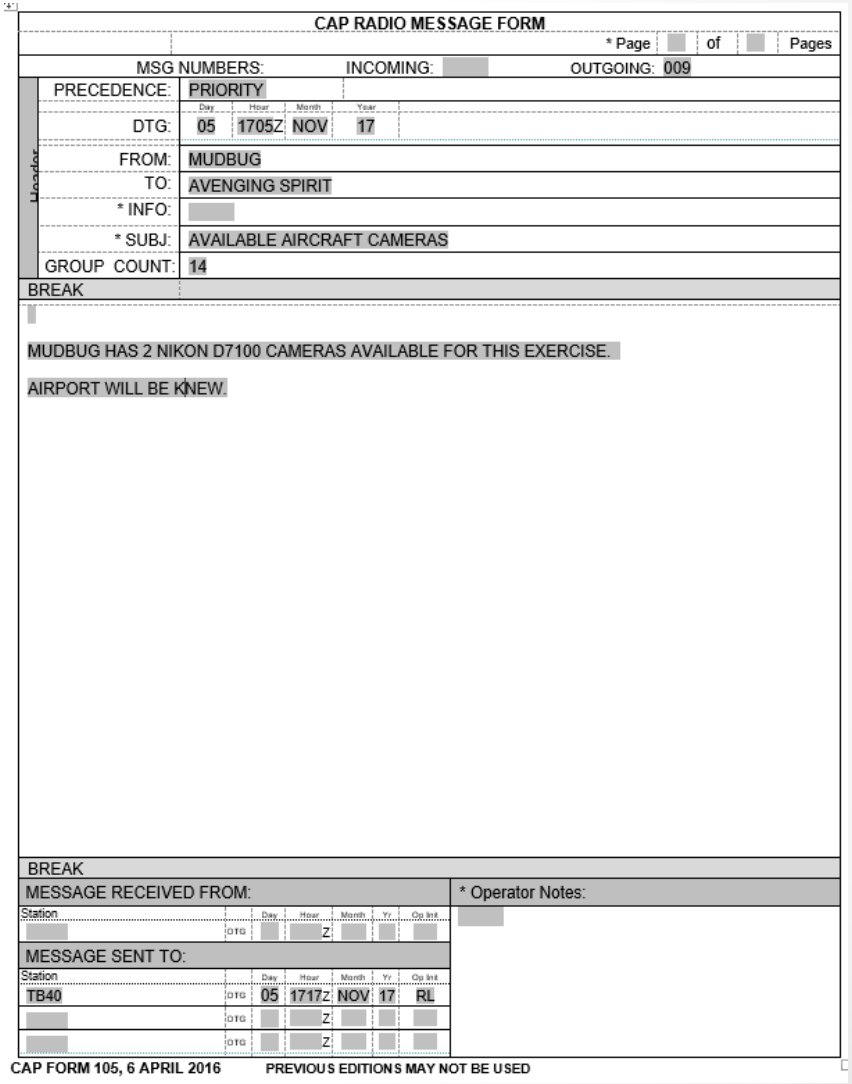
Station		Day	Hour	Month	Yr	Op Init
TB33/40	DTG	05	1640Z	NOV	17	RL

MESSAGE SENT TO:

Station		Day	Hour	Month	Yr	Op Init
	DTG		Z			
	DTG		Z			
	DTG		Z			

* Operator Notes:

CAP FORM 105, 6 APRIL 2016 PREVIOUS EDITIONS MAY NOT BE USED



CAP RADIO MESSAGE FORM

* Page of Pages

MSG NUMBERS: INCOMING: OUTGOING: 009

Header

PRECEDENCE:	PRIORITY
DTG:	Day 05 Hour 1705Z Month NOV Year 17
FROM:	MUDBUG
TO:	AVENGING SPIRIT
* INFO:	
* SUBJ:	AVAILABLE AIRCRAFT CAMERAS
GROUP COUNT:	14

BREAK

MUDBUG HAS 2 NIKON D7100 CAMERAS AVAILABLE FOR THIS EXERCISE.

AIRPORT WILL BE KNEW.

BREAK

MESSAGE RECEIVED FROM:

Station		Day	Hour	Month	Yr	Op Init
	DTG		Z			

MESSAGE SENT TO:

Station		Day	Hour	Month	Yr	Op Init
TB40	DTG	05	1717Z	NOV	17	RL
	DTG		Z			
	DTG		Z			

* Operator Notes:

CAP FORM 105, 6 APRIL 2016 PREVIOUS EDITIONS MAY NOT BE USED

Prohibited Operating Practices

- Violation of Radio Silence
- Personal Conversation
- Transmitting in a Net without permission of NCS
- Lack of identifying call sign
- Excessive tuning and testing
- Use of Amateur Radio or Citizens Band frequencies for CAP business, and Vice-Versa.
- Use of 10 codes or Amateur Radio Q Signals
- Use of individual's names or release of personal identification information

Floor Demos / Practical

- Person to person relay – simple
  - "blue dog"
- Person to person relay – not simple
  - "GT1 is located at the What-A-Burger 789 Jonas road 1.3 miles past Highway 6"
- Message form Team 1
- Message form Team 2

ICUT EVAL REQUIREMENTS

RADIO OFF

- Check AC power cord and antenna cable
- Turn the radio on
- Adjust the volume
- Change channel – at least two different zone and channel combinations
- Adjust Hi/Low power
- Turn scan on and off
- Set squelch to "squelch select"
- Instructor Demo: Accidental "Scan ED", "C/S" or "KY SEL" procedure.

ICUT EVAL REQUIREMENTS

RADIO ON

- Turn off radio
- Connect Microphone (If applicable)
- Demonstrate holding micro- phone 2-3 in. from mouth

ICUT EVAL REQUIREMENTS

OPS

- Send informal tactical message successfully
- Who will the message go to? (Importance of delivery)
- When are MAYDAY SECURITE, and PAN-PAN used?
- Give at least 2 of 10 prohibited practices
- Define tactical call signs, functional designators, & when each are used
- Define "Directed Net" and "Free Net"

Johnson VHF Mobile



https://www.youtube.com/watch?v=sgjsTIwLbZU

Johnson VHF Mobile



Change Zone and Channel

- Zone and Channel are found on the right side of the display
- Zone is the number on left
- Channel is number on right
- Change the Zone and Channel using the Select knob below them



Change Zone and Channel

- Change the Channel by pressing the Select button to position the bar above the channel



Johnson VHF HT

5100 ES

7/800 MHz | VHF | UHF

Featuring AES encryption and the AMBE+2 vocoder which provides loud and clear audio, the 5100 ES portable radio is durable and interoperable.



VHF Freq Designators

ZONE 1		
TACTICAL		
Channel	**Title**	**Definition**
1	CC1	Command and control
2	CC2	Command and control
3	AIR 1	Air to Air
4	AIR 2	Air to Air
5	GUARD 1	Emergency/Hailing
6	TAC 1	Tactical Ch 1
7	R65 CAT	Rec Only ODD# Repeaters
8	R66 CAT	Rec Only EVEN# Repeaters
9	R67 ABNCH3	Airborne repeater channel 3
10	R68 ABNCH 4	Airborne repeater channel 4
11	R69 ABNCH5	Airborne repeater channel 5
12	R70 GND CH12	Ground Repeater channel 12
13	R63 GND CH 7	Ground Repeater channel 7
14	R64 GND CH 8	Ground Repeater Channel 8

ZONE 2		
LAWGARPTRS		
Channel	**Title**	**Definition**
1	ALEXANDRIA	Alexandria analog repeater
2	Baton Roug	Baton Rouge analog repeater
3	Houma	Houma analog repeater
4	JENNINGS	Jennings analog repeater
5	LACOMBE	Lacombe analog repeater
6	LAFAYETTE	Lafayette analog repeater
7	LAKE CHARL	Lake Charles Analog Repeater
8	MONROE	Monroe analog repeater
9	MORGANCITY	Morgan City analog repeater
10	NEWORLEANS	New Orleans analog repeater
11	SHREVEPORT	Shreveport analog repeater

Other Zones

Zone 3 – Digital Repeaters

Zone 5 – Weather

Zone 7 – Liasion channels

Zone 16 – Digital Tactical

HF Micom3 wALE





Van HF Antenna



Ground Sortie

Establish a forward communication base in the field with both VHF and HF capability

- **Challenge**
  - **Determine - Suitable location**
  - **Determine - Source of power**
    - **Batteries, generator, power grid**
  - **Assemble the comm post**
  - **Fly the antennas**
  - **Operate the Station**

ES Signoffs Covered
L-0001
L-0002
L-0003
L-0005
L-0014

HF RDP
Rapid Deployment Package



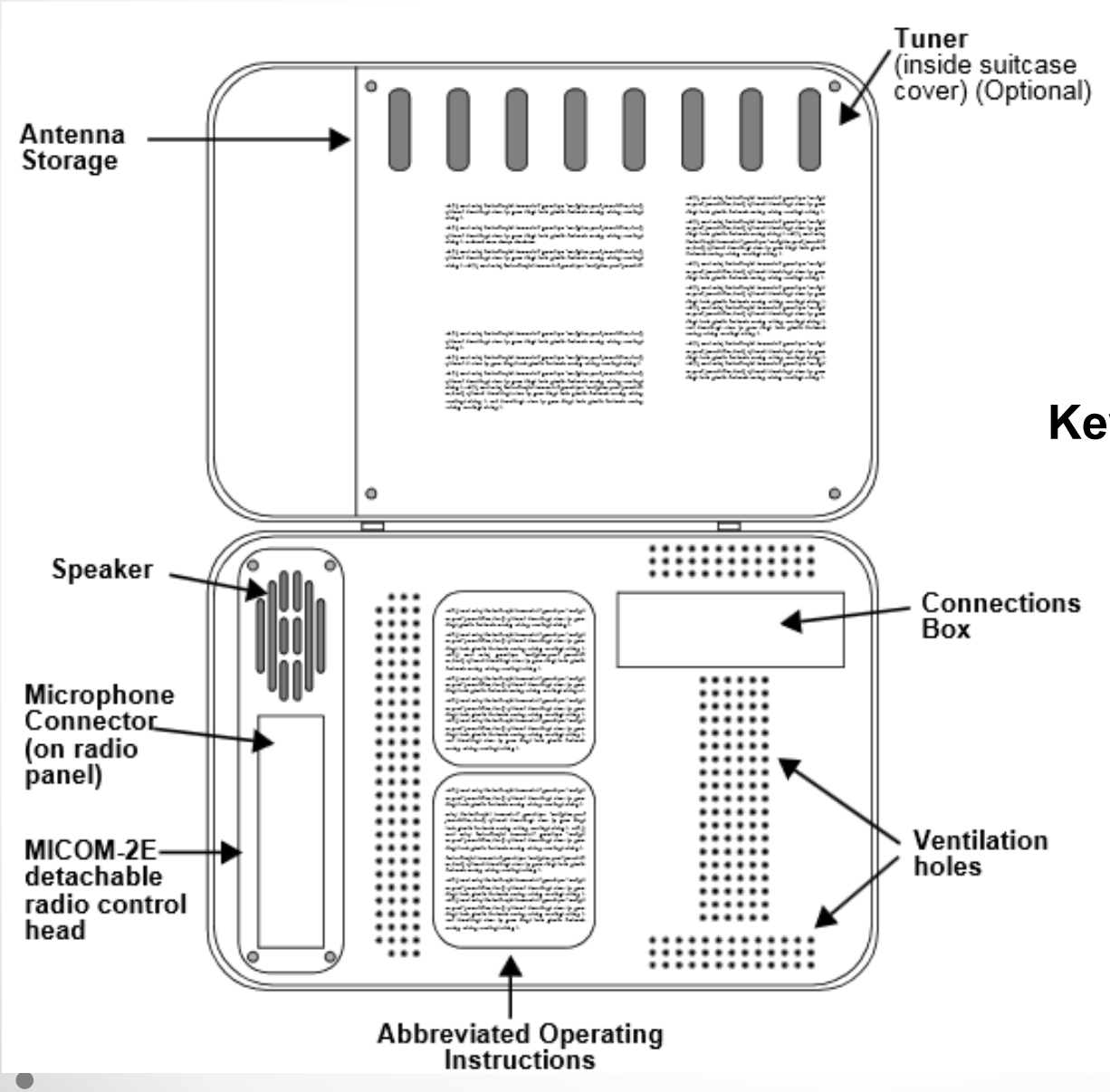
Complete HF radio system in a “Go Kit”

Includes

- **HF radio – with ALE**
- **Antenna Tuner**
- **Antenna**
- **Cables & Ground rods**
- **Power supplies – AC or DC**

Just set up and add power

HF RDP - Radio Parts

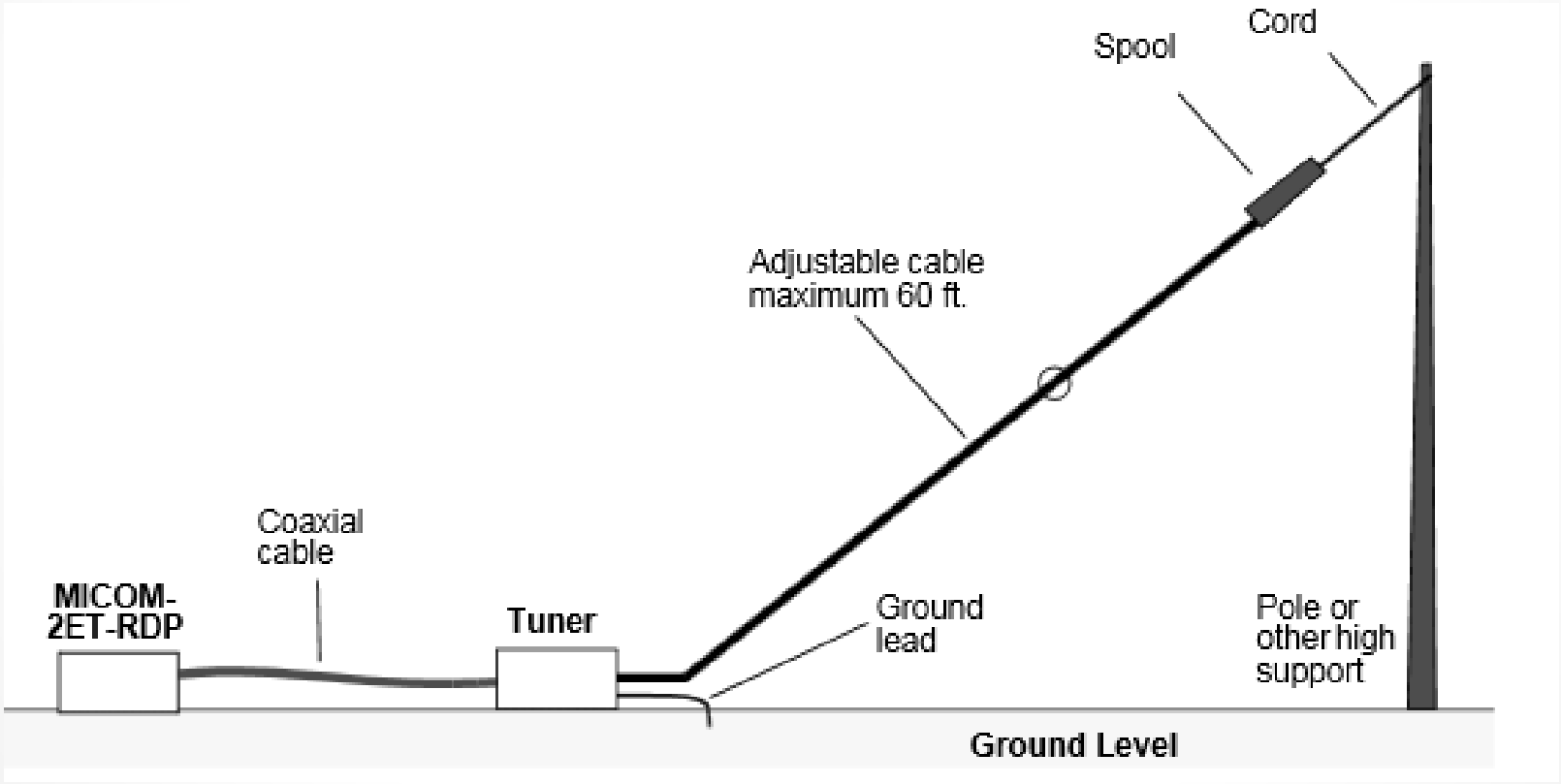


Key Parts

- **Radio**
- **Antenna Tuner**
- **Antenna**
- **AC/DC power supply**

HF RDP - Antenna

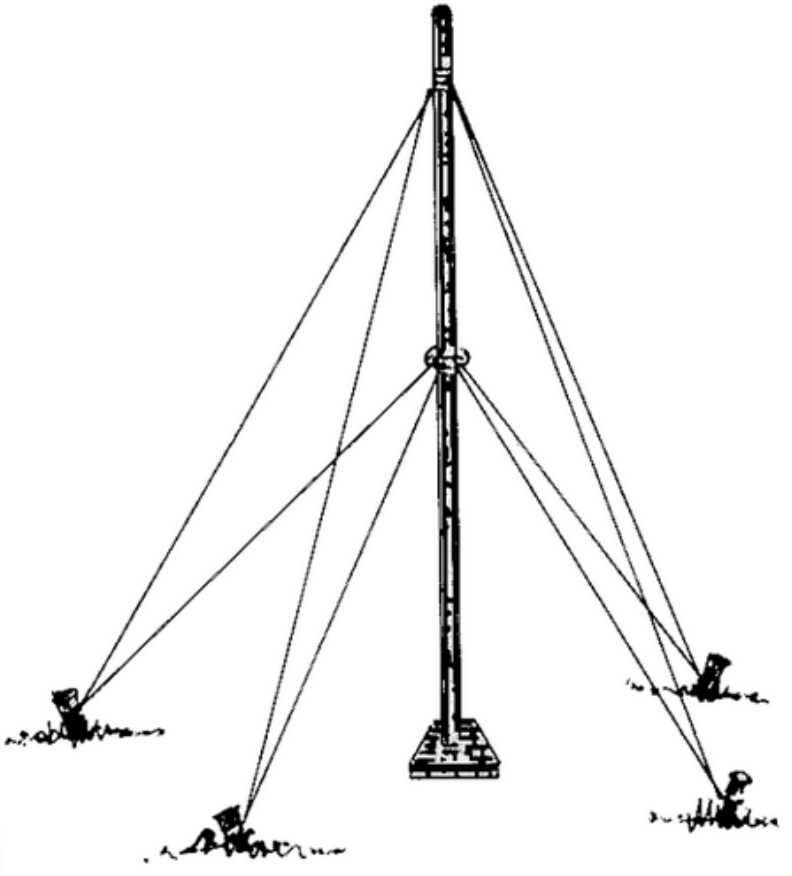
RDP - Sloping Long Wire Antenna configuration



AB-155 Antenna Mast

40 ft Antenna Mast that assembles on ground and tilt/lifted into vertical position





Also referred to as a GRV-4

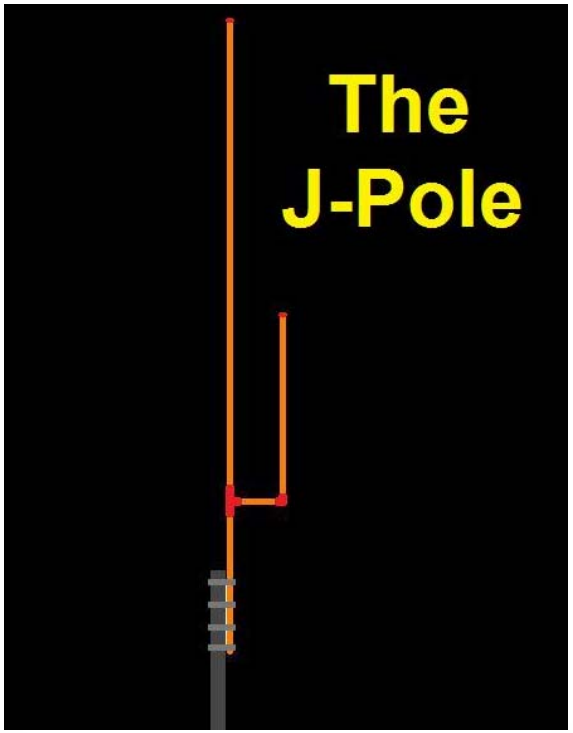
VHF Antennas

For Comm Base – Get the antenna as high as possible

Mag Mount



J-Pole Base



J-Pole Flexiable

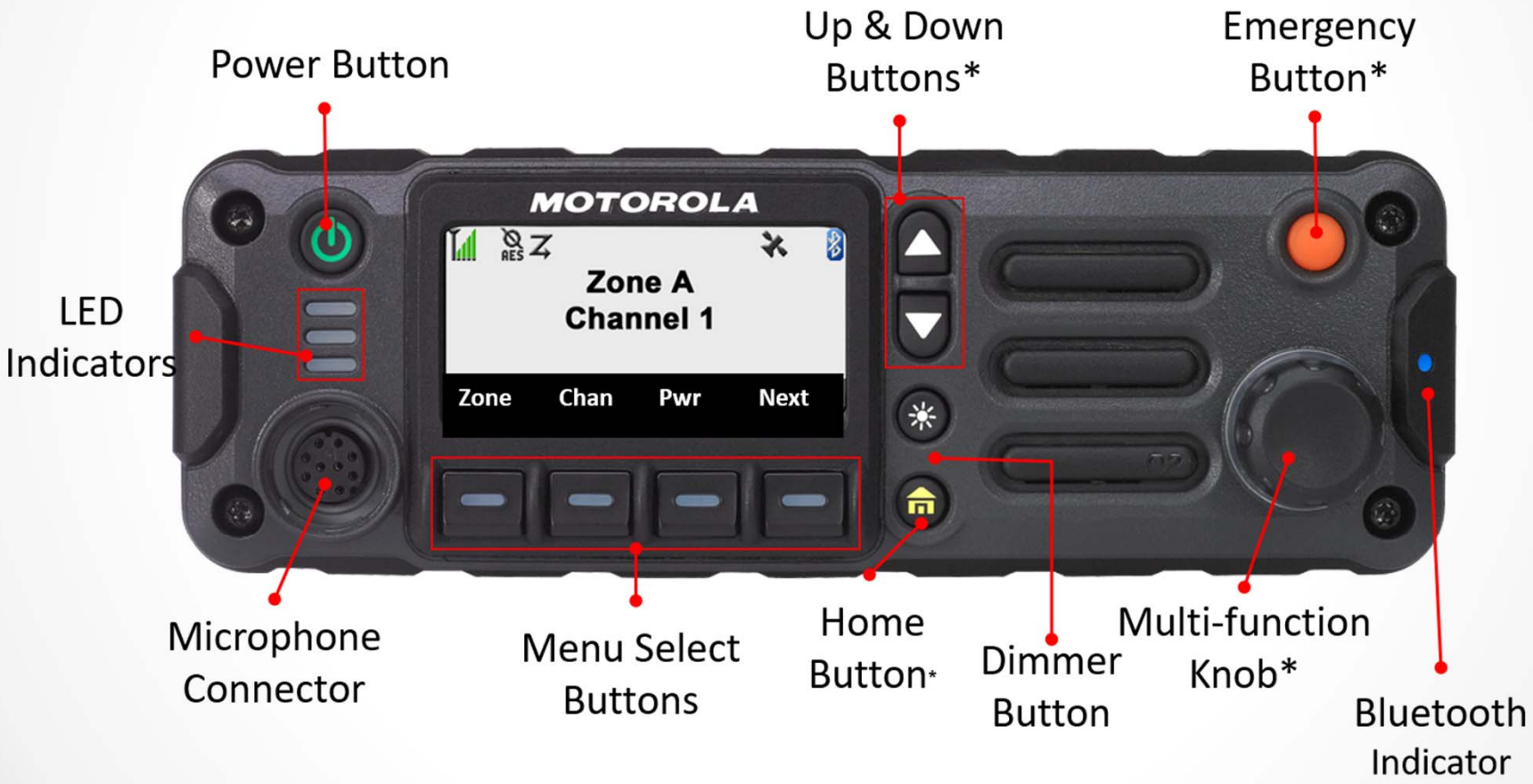


Motorola APX 4500

VHF RADIO BASE/MOBILE UNITS

Key Items to Know

APX 4500 Front



*** These radio controls/buttons are programmable.**

Radio On/Off

- Turn On
  - Press The Power button briefly
  - Red/Yel/Green will light
- Turn Off
  - Press the Power button



WARNING – This radio talks to you!

Volume Control

- Multi-Function knob
  - Use to adjust volume
  - Display will show volume in bars when adjusting
- Raise Volume
  - MF Knob clockwise
- Lower Volume
  - MF Knob counterclockwise



Change Zones

- To Change Zones
  - Depress the left “Menu Select Button” shown as “Zone”
  - Use “Up & Down” buttons to change the Zone
  - When you have the desired Zone….
  - Press the left “Menu Select Button” again to make the selection.

    NOTE – the button display shows “SEL” for select



Change Channels

- To Change Channel
  - Depress the 2nd from left “Menu Select Button” shown as “CHANNEL”
  - Use “Up & Down” buttons to change the Channel
  - When you have the desired Channel….
  - Press the 2nd from left “Menu Select Button” again to make the selection.

    NOTE – the button display shows “SEL” for select



Change Zone/Channels Alt

- Multi-Function knob
  - Push in the MF knob
  - Green ring will lite for 5 seconds
- Change Channel
  - Rotate the MF Knob to desired channel
- Change Zone
  - Use the UP/Down buttons to change the zone
- MF knob
  - Returns to Volume Control after 5 seconds



APX 4500
Channel Matrix

- National Channel plug is installed on all CAP radios.
  - Zone 1&2 are the Simplex channels
  - Zones 3-10 are the repeaters (All Wings)

- Additionally, Louisiana Repeaters are grouped together into a single zone.
  - Zone 17 – LAWG Analog Repeaters
  - Zone 18 – LAWG Digital Repeaters

- Handheld APX radios use the same channel matrix

APX 4500

Channel Matrix - 1

MOTOROLA APX 4500 -- CHANNEL MATRIX

ZONE 1 TACTICAL			ZONE 2 P25 ZN1			ZONE 17 LAWG RPTR ANALOG			ZONE 18 LAWG RPTR DIGITAL			ZONE 3 R01-R16			ZONE 4 R01P-R16P			ZONE 5 R17-R32			ZONE 6 R17P-R32P			ZONE 7 R33-R48			ZONE 8 R33P-R48P		
1	CC1	Command and control	1	CC1P	Digital/P25 SIMPLEX	1	ALEXAND R19	Alexandria analog repeater	1	ALEXAND R19P	Alexandria Digital repeater	1	R01	Repeater	1	R01P	Digital/P25 repeater	1	R17	Repeater	1	R17P	Digital/P25 repeater	1	R33	Repeater	1	R33P	Digital/P25 repeater
2	CC2	Command and control	2	CC2P	Digital/P25 SIMPLEX	2	BATON ROUGE R31	Baton Rouge analog repeater	2	BATON ROUG R31P	Baton Rouge Digital repeater	2	R02	Repeater	2	R02P	Digital/P25 repeater	2	R18	Repeater	2	R18P	Digital/P25 repeater	2	R34	Repeater	2	R34P	Digital/P25 repeater
3	AIR 1	Air to Air	3	AIR1P	Digital/P25 SIMPLEX	3	HOUMA R18	Houma analog repeater	3	HOUMA R18P	Houma Digital repeater	3	RO3	Repeater	3	R03P	Digital/P25 repeater	3	R19	Repeater	3	R19P	Digital/P25 repeater	3	R35	Repeater	3	R35P	Digital/P25 repeater
4	AIR 2	Air to Air	4	AIR2P	Digital/P25 SIMPLEX	4	JENNINGS R11	Jennings analog repeater	4	JENNINGS R11P	Jennings Digital repeater	4	RO4	Repeater	4	R04P	Digital/P25 repeater	4	R20	Repeater	4	R20P	Digital/P25 repeater	4	R36	Repeater	4	R36P	Digital/P25 repeater
5	GUARD 1	Emergency/Hailing	5	GUARD1P	Digital/P25 SIMPLEX	5	LACOMB R56	Lacombe analog repeater	5	LACOMB R56P	Lacombe Digital repeater	5	RO5	Repeater	5	R05P	Digital/P25 repeater	5	R21	Repeater	5	R21P	Digital/P25 repeater	5	R37	Repeater	5	R37P	Digital/P25 repeater
6	TAC 1	Tactical Ch 1	6	TAC1P	Digital/P25 SIMPLEX	6	LAFAYETT R44	Lafayette analog repeater	6	LAFAYETT R44P	Lafayette Digital repeater	6	RO6	Repeater	6	R06P	Digital/P25 repeater	6	R22	Repeater	6	R22P	Digital/P25 repeater	6	R38	Repeater	6	R38P	Digital/P25 repeater
7	Chan 1 Mon		7	Chan 1 MonP		7	LAKE CHARL R36	Lake Charles Analog Repeater	7	LAKE CHAR R36P	Lake Charles Digital Repeater	7	RO7	Repeater	7	R07P	Digital/P25 repeater	7	R23	Repeater	7	R23P	Digital/P25 repeater	7	R39	Repeater	7	R39P	Digital/P25 repeater
8	Chan 2 Mon		8	Chan 2 MonP		8	MONROE R40	Monroe analog repeater	8	MONROE R40P	Monroe Digital repeater	8	RO8	Repeater	8	R08P	Digital/P25 repeater	8	R24	Repeater	8	R24P	Digital/P25 repeater	8	R40	Repeater	8	R40P	Digital/P25 repeater
9	TAC Rptr Mon		9	TAC Rptr MonP		9	MORGAN CITY R33	Morgan City analog repeater	9	MORGAN CITY R33P	Morgan City Digital repeater	9	RO9	Repeater	9	R09P	Digital/P25 repeater	9	R25	Repeater	9	R25P	Digital/P25 repeater	9	R41	Repeater	9	R41P	Digital/P25 repeater
10	R67	Airborne repeater channel 3	10	R67P		10	NEW ORL R25	New Orleans analog repeater	10	NEW ORL R25P	New Orleans Digital repeater	10	RO10	Repeater	10	R010P	Digital/P25 repeater	10	R26	Repeater	10	R26P	Digital/P25 repeater	10	R42	Repeater	10	R42P	Digital/P25 repeater
11	R68	Airborne repeater channel 4	11	R68P		11	SHRVPORT R17	Shreveport analog repeater	11	SHRVPORT R17P	Shreveport Digital repeater	11	RO11	Repeater	11	R011P	Digital/P25 repeater	11	R27	Repeater	11	R27P	Digital/P25 repeater	11	R43	Repeater	11	R43P	Digital/P25 repeater
12	R69	Airborne repeater channel 5	12	R69P		12			12			12	RO12	Repeater	12	R012P	Digital/P25 repeater	12	R28	Repeater	12	R28P	Digital/P25 repeater	12	R44	Repeater	12	R44P	Digital/P25 repeater
13	R70	Ground Repeater channel 12	13	R70P		13			13			13	RO13	Repeater	13	R013P	Digital/P25 repeater	13	R29	Repeater	13	R29P	Digital/P25 repeater	13	R45	Repeater	13	R45P	Digital/P25 repeater
14	R63	Ground Repeater channel 7	14	R63P		14			14			14	RO14	Repeater	14	R014P	Digital/P25 repeater	14	R30	Repeater	14	R30P	Digital/P25 repeater	14	R46	Repeater	14	R46P	Digital/P25 repeater
15	R64	Ground Repeater Channel 8	15	R64P		15			15			15	RO15	Repeater	15	R015P	Digital/P25 repeater	15	R31	Repeater	15	R31P	Digital/P25 repeater	15	R47	Repeater	15	R47P	Digital/P25 repeater
16	N/A		16			16			16			16	RO16	Repeater	16	R016P	Digital/P25 repeater	16	R32	Repeater	16	R32P	Digital/P25 repeater	16	R48	Repeater	16	R48P	Digital/P25 repeater

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Channel Matrix - 2

ZONE 9 R49-R64			ZONE 10 R49P-R64P			ZONE 11 BORDER ZONE			ZONE 12 USCG INTEROP			ZONE 13 VTAC INTEROP			ZONE 14 NTIA INTEROP			ZONE 15 NOAA WX			ZONE 16 RESERVED			ZONE X			ZONE X		
1	R49	Repeater	1	R49P	Digital/P25 repeater	1	R101		1	CG 06	Narrowband 156.300 Simplex	1	VCALL 10		1	NC 1		1	WX 1	162.55	1			1			1		
2	R50	Repeater	2	R50P	Digital/P25 repeater	2	R102		2	CG 16	Narrowband 156.800 simplex (HAILER)	2	VTAC 11		2	IR 1	RPTR	2	WX 2	162.4	2			2			2		
3	R51	Repeater	3	R51P	Digital/P25 repeater	3	R103		3	CG 21A	Narrowband 157.0500 simplex	3	VTAC 12		3	IR 2	RPTR	3	WX 3	162.475	3			3			3		
4	R52	Repeater	4	R52P	Digital/P25 repeater	4	R104		4	CG22A	Narrowband 157.100 simplex	4	VTAC 13		4	IR 3	RPTR	4	WX 4	162.425	4			4			4		
5	R53	Repeater	5	R53P	Digital/P25 repeater	5	R105		5	CG23A	Narrowband 157.150 simplex	5	VTAC 14		5	IR 4	RPTR	5	WX 5	162.45	5			5			5		
6	R54	Repeater	6	R54P	Digital/P25 repeater	6	R106		6	CG81A	Narrowband 157.0750 simplex	6	VTAC 17		6	IR 5	SIMPLEX	6	WX 6	162.5	6			6			6		
7	R55	Repeater	7	R55P	Digital/P25 repeater	7	R107		7	CG82A	Narrowband 157.1250 simplex	7	VTAC 17D		7	IR 6	SIMPLEX	7	WX 7	162.525	7			7			7		
8	R56	Repeater	8	R56P	Digital/P25 repeater	8	R101P		8	CG83A	Narrowband 157.1750 simplex	8	VTAC 33		8	IR 7	SIMPLEX	8	WX 8	161.65	8			8			8		
9	R57	Repeater	9	R57P	Digital/P25 repeater	9	R102P		9			9	VTAC 34		9	IR 8	SIMPLEX	9	WX 9	161.775	9			9			9		
10	R58	Repeater	10	R58P	Digital/P25 repeater	10	R103P		10			10	VTAC 35		10	IR 9	SIMPLEX	10	WX 10	163.275	10			10			10		
11	R59	Repeater	11	R59P	Digital/P25 repeater	11	R104P		11			11	VTAC 36		11			11			11			11			11		
12	R60	Repeater	12	R60P	Digital/P25 repeater	12	R105P		12			12	VTAC 37		12			12			12			12			12		
13	R61	Repeater	13	R61P	Digital/P25 repeater	13	R106P		13			13	VTAC 38		13			13			13			13			13		
14	R62	Repeater	14	R62P	Digital/P25 repeater	14	R107P		14			14	VSAR 16		14			14			14			14			14		
15	R63	Repeater	15	R63P	Digital/P25 repeater	15	R110		15			15			15			15			15			15			15		
16	R64	Repeater	16	R64P	Digital/P25 repeater	16	R111		16			16			16			16			16			16			16		

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Channel Matrix - LAWG

ZONE 1 TACTICAL			ZONE 2 P25 ZN1			ZONE 17 LAWG RPTR ANALOG			ZONE 18 LAWG RPTR DIGITAL		
1	CC1	Command and control	1	CC1P	Digital/P25 SIMPLEX	1	ALEXAND R19	Alexandria analog repeater	1	ALEXAND R19P	Alexandria Digital repeater
2	CC2	Command and control	2	CC2P	Digital/P25 SIMPLEX	2	BATON ROUGE R31	Baton Rouge analog repeater	2	BATON ROUG R31P	Baton Rouge Digital repeater
3	AIR 1	Air to Air	3	AIR1P	Digital/P25 SIMPLEX	3	HOUMA R18	Houma analog repeater	3	HOUMA R18P	Houma Digital repeater
4	AIR 2	Air to Air	4	AIR2P	Digital/P25 SIMPLEX	4	JENNINGS R11	Jennings analog repeater	4	JENNINGS R11P	Jennings Digital repeater
5	GUARD 1	Emergency/Hailing	5	GUARD1P	Digital/P25 SIMPLEX	5	LACOMB R56	Lacombe analog repeater	5	LACOMB R56P	Lacombe Digital repeater
6	TAC 1	Tactical Ch 1	6	TAC1P	Digital/P25 SIMPLEX	6	LAFAYETT R44	Lafayette analog repeater	6	LAFAYETT R44P	Lafayette Digital repeater
7	Chan 1 Mon		7	Chan 1 MonP		7	LAKE CHARL R36	Lake Charles Analog Repeater	7	LAKE CHAR R36P	Lake Charles Digital Repeater
8	Chan 2 Mon		8	Chan 2 MonP		8	MONROE R40	Monroe analog repeater	8	MONROE R40P	Monroe Digital repeater
9	TAC Rptr Mon		9	TAC Rptr MonP		9	MORGAN CITY R33	Morgan City analog repeater	9	MORGAN CITY R33P	Morgan City Digital repeater
10	R67	Airborne repeater channel 3	10	R67P		10	NEW ORL R25	New Orleans analog repeater	10	NEW ORL R25P	New Orleans Digital repeater
11	R68	Airborne repeater channel 4	11	R68P		11	SHRVPORT R17	Shreveport analog repeater	11	SHRVPORT R17P	Shreveport Digital repeater
12	R69	Airborne repeater channel 5	12	R69P		12			12		
13	R70	Ground Repeater channel 12	13	R70P		13			13		
14	R63	Ground Repeater channel 7	14	R63P		14			14		
15	R64	Ground Repeater Channel 8	15	R64P		15			15		
16	N/A		16			16			16		

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LAWG Repeater Cheat Sheet

LAWG REPEATERS	
ALEXANDRIA R19	R19
BATON ROUGE R31	R31
HOUMA R18	R18
JENNINGS R11	R11
LACOMB R56	R56
LAFAYETTE R44	R44
LAKE CHARLES R36	R36
MONROE R40	R40
MORGAN CITY R33	R33
NEW ORLEANS R25	R25
SHREVEPORT R17	R17

ZONE 5 R17-R32		
1	R17	Repeater
2	R18	Repeater
3	R19	Repeater
4	R20	Repeater
5	R21	Repeater
6	R22	Repeater
7	R23	Repeater
8	R24	Repeater
9	R25	Repeater
10	R26	Repeater
11	R27	Repeater
12	R28	Repeater
13	R29	Repeater
14	R30	Repeater
15	R31	Repeater
16	R32	Repeater

ZONE 9 R49-R64		
1	R49	Repeater
2	R50	Repeater
3	R51	Repeater
4	R52	Repeater
5	R53	Repeater
6	R54	Repeater
7	R55	Repeater
8	R56	Repeater
9	R57	Repeater
10	R58	Repeater
11	R59	Repeater
12	R60	Repeater
13	R61	Repeater
14	R62	Repeater
15	R63	Repeater
16	R64	Repeater

APX 4000 Handheld



Display Soft keys

Up/Down Selector



Has the same Zone/Channel Matrix as the APX 4500s

APX 4000 Zone/Channel



Display
Soft keys

Up/Down
Selector

Channel
Selector

On/Off
Volume



Change Zone

- **Use Display soft key and Up/Down selector**

Change Channel

- **Use Display soft key and Up/Down selector**
- **Or Top knob channel selector**